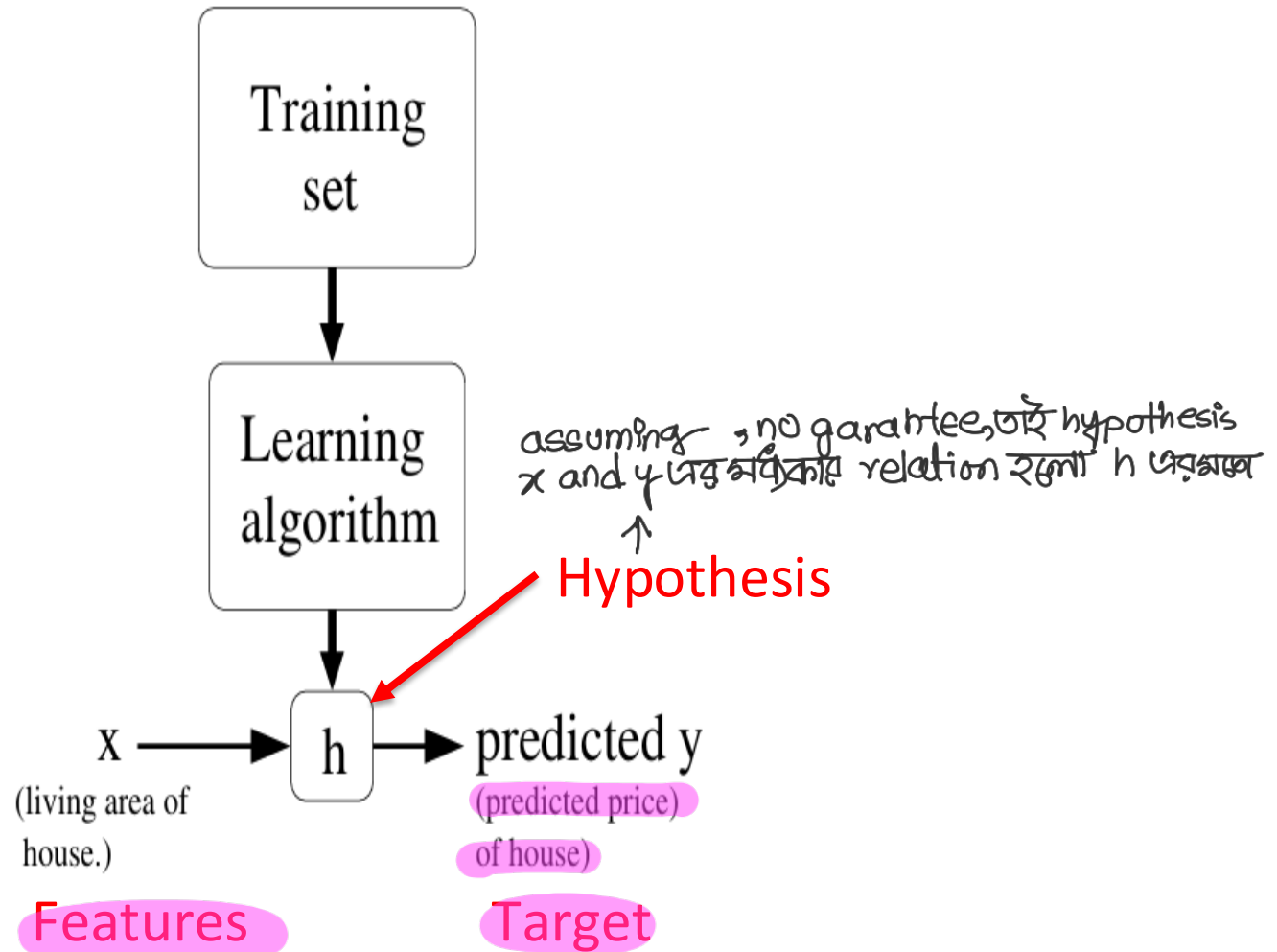


Linear Regression

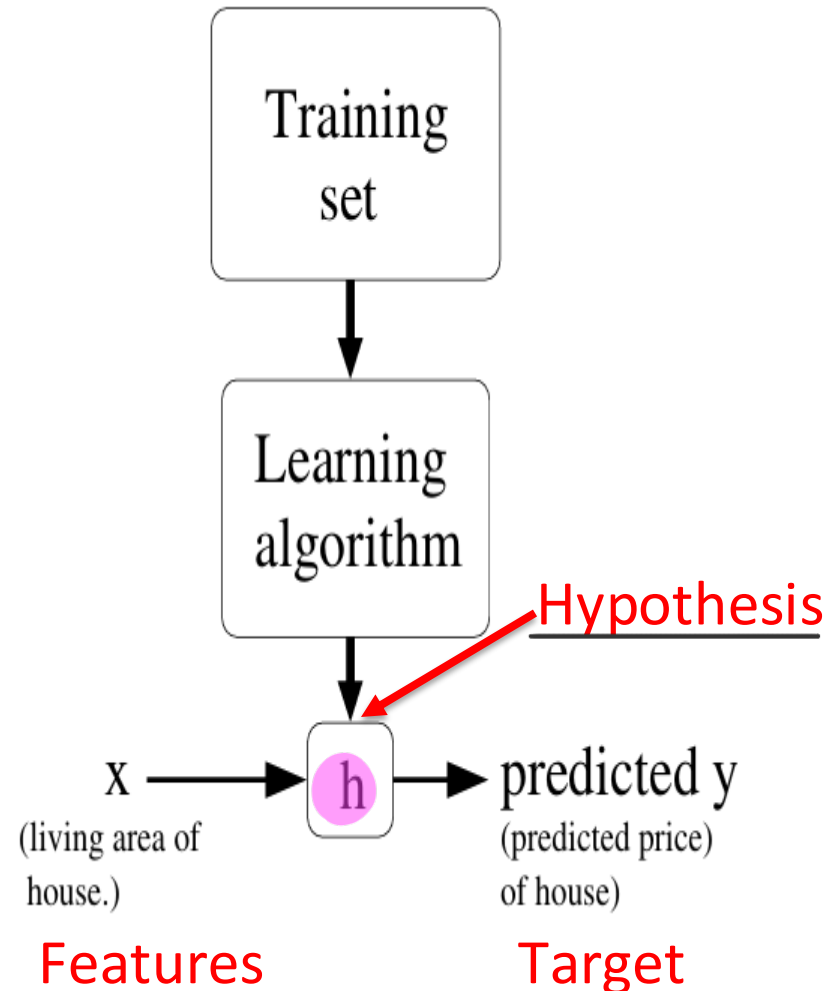
যেসকলো continuous data তক predict করা,
x দেওয়া থাকলে y predict করতে হবে, So এসকলো
range হ y continuous.

Supervised Learning (Recap)



Supervised Learning

- $x^{(i)}$: input or feature
- $y^{(i)}$: output or target
- $(x^{(i)}, y^{(i)})$: i^{th} training example
- $\{(x^{(i)}, y^{(i)}): i = 1, 2, \dots, n\}$ the training set
- Goal??



Supervised Learning

- $\mathbf{X}$ is the input space ; $x \rightarrow \text{input}$
- $\mathbf{y}$ is the target space ; $y \rightarrow \text{output/target}$

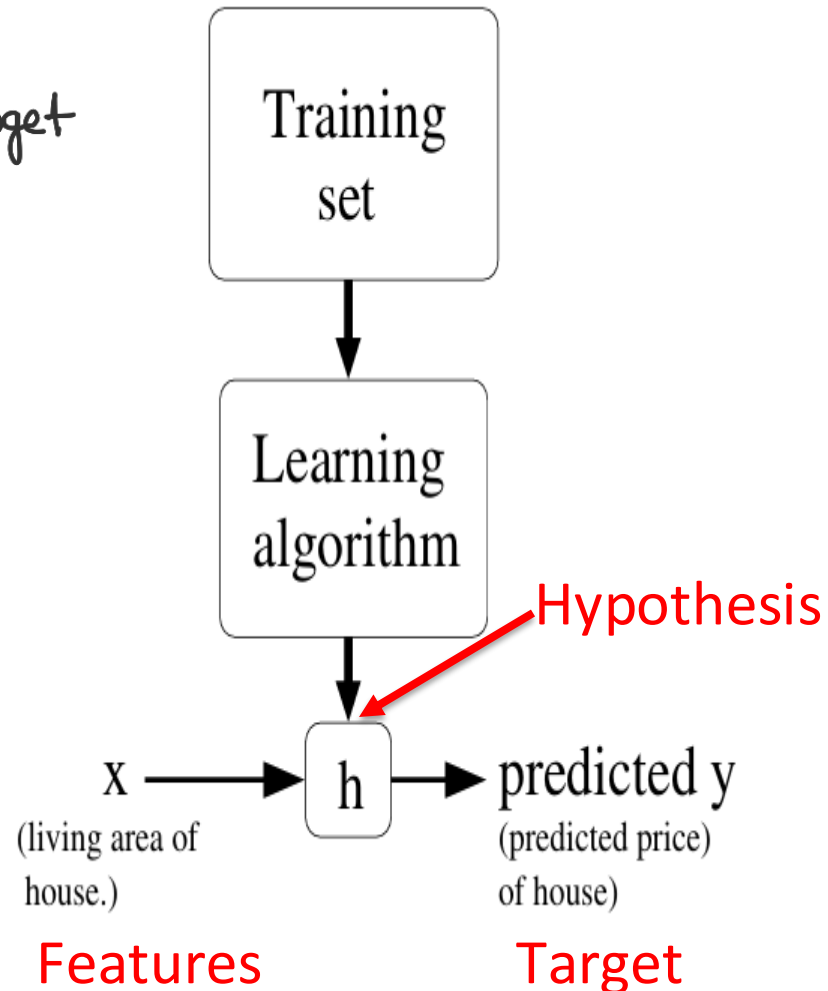
Goal:

- Learn a **function** or **hypothesis** such that

$$h: X \rightarrow Y$$

i.e.

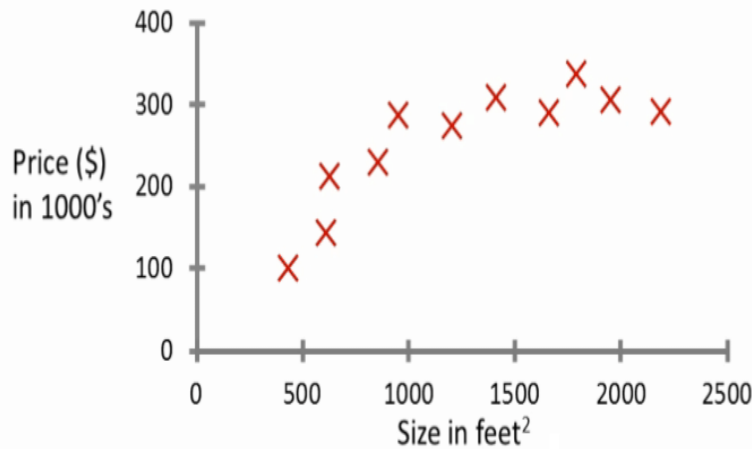

$$h(x) = y$$



Housing Price Prediction

Regression !!!

Housing price prediction.



(1 feature given)	
Living area (feet ²)	Price (1000\$)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

Learn a function h such that

$$h(\text{living area in } \textit{feet}^2) \sim \text{Price in } 1000\$s$$

Living area as price predictor

approximate,
we don't know if it exactly or not

Regression vs Classification

- Regression
 - Target variable is continuous
 - Example,
 - Housing price prediction
- Classification
 - Target variable can take only small number of discrete values
 - Example,
 - Email classification (Spam vs Ham)

Linear Regression

we assume, x and y are relation linear.

- We assume a linear relation between X and Y .
- Given x , we approximate y with a linear function.
- Housing price prediction with two features

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

→ এটা না থাকলে তো সব line origin দিয়ে যাবে।

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

$$x_0 = 1$$

- θ is called parameters/weights

- A $d+1$ dimensional vector of weights

Living area (feet ²)	#bedrooms	Price (1000\$)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
$\vdots$	$\vdots$	$\vdots$

→ assuming
→ transpose কারণ এটা সারি
column vector চিত্রাকর্ষিত,
to multiply with x .
→ $(d+1)$ dimensional এর x vector.

Linear Regression

- Goal:
 - Given a training set, learn θ
 - Such that $h_{\theta}(x)$ is as close as possible to y

আমাদের এমন h_{θ} বের করতে হবে যাতে $h_{\theta}(x) \approx y$ (y predicted value এর as close as possible হয়)

- If we had only one sample, minimize

$$J(\theta) = \frac{1}{2} (h_{\theta}(x) - y)^2$$

($\rightarrow$ যে কোন সমস্যা, $(-)$ or $(+)$ we use square. So we use square.)

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

Linear Regression

- Goal:
 - Given a training set, learn θ
 - Such that $h_{\theta}(x)$ is as close as possible to y

Least ~~Mean~~ Squares (LMS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2} \sum_{k=1}^n \left(h_{\theta}(x^{(k)}) - y^{(k)} \right)^2$$

Handwritten notes: "sum of the all samples" with an arrow pointing to the summation index n ; "[Least square (LS)]" in blue.

Cost Function/Loss Function/
objective function

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = \operatorname{argmin}_{\theta} J(\theta)$$

- How?

Linear Regression

$$J(\theta) = \frac{1}{2} \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)})^2$$

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = \operatorname{argmin}_{\theta} J(\theta)$$

- Set partial derivatives to 0 to obtain minima

$$\frac{\partial J}{\partial \theta_0} = \sum_{k=1}^n (\underbrace{\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)}}_{1} - y^{(k)}) \cdot \underbrace{1}_{1} = 0$$

$$\frac{\partial J}{\partial \theta_1} = \sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) \underline{x_1^{(k)}} = 0$$

$$\frac{\partial J}{\partial \theta_2} = \sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) \underline{x_2^{(k)}} = 0$$

Basically equation
গুণিতা যে parabola,
তাই সেই lowest
value। তাই minima
নয়, not maxima

Linear Regression

- Simplifying the partial derivative equations,

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) = 0$$

$$\Rightarrow n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_1^{(k)} = 0$$

$$\Rightarrow \theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_2^{(k)} = 0$$

$$\Rightarrow \theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

System of Linear Equations

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- Vectorized Form

$$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- Vectorized form

$$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

$$(X^T X)\theta = X^T y$$

X is a $n \times (d + 1)$ matrix
Each row is a multivariate sample

y is a $n \times 1$ vector
Each row is a label of a sample

$$X = \begin{bmatrix} x_0 & x_1 & x_2 \\ [1 & 5 & 10] \\ [1 & 20 & 25] \\ [1 & 30 & 50] \\ [1 & 100 & 250] \end{bmatrix}$$

$n \times (d+1)$
 n : no. of rows $\rightarrow$ data points
 d : no. of features

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ 1 & x_{31} & x_{32} \\ 1 & x_{41} & x_{42} \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ x_{11} & x_{21} & x_{31} & x_{41} \\ x_{12} & x_{22} & x_{32} & x_{42} \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ 1 & x_{31} & x_{32} \\ 1 & x_{41} & x_{42} \end{bmatrix} = \begin{bmatrix} \eta & x_1^{(k)} \end{bmatrix}$$

Linear Regression

- Solve the system of linear equations to get θ^*

$$\underline{(X^T X)\theta = X^T y}$$

$$\theta^* = \underline{(X^T X)^{-1} X^T y}$$

Normal Equation

Linear Regression

- Once trained predict housing prices with,

$$\hat{y} = \cancel{\theta^T x} \quad (\theta^*)^T x$$

- What if $n \gg 1$?? *time-consuming (n large value অনেক বড় হলে matrix অনেক বড় হয়)*
- What if $d \gg 1$?? *প্রত্যেকটি feature related each equation থাকবে, so equation হবে অনেক*
- What if the $(X^T X)$ is non-invertible? *singular matrix $\rightarrow$ row = 2, column = 2 (যদি কোন sample একই হয় তাহলে একই row = 2, column = 2) (যদি কোন sample একই হয় তাহলে একই row = 2, column = 2)*
- What if the relation between X and y are not linear? *gradient descent* *data হবে তার নিজস্ব (data হবে তার নিজস্ব)*

$X^T X$ বড় করে time consuming

$\rightarrow$ এই সমস্যাটি problem solve করতে we need gradient descent.

Reference

- Dive into Deep Learning (d2l)
 - Sections 3.1 (up to 3.1.2), 3.4
- Deep Learning [Optional Reading]
 - Section 5.1.4